

Module 3 Homework Exception Control

 Published

Assignment: Exception Control in Java

Objective: The objective of this assignment is to assess your understanding of exception handling in Java.

Instructions: Write a Java program that demonstrates the use of exception handling to handle various types of exceptions. Your program should include at least two custom exception classes that you create.

1. **Exception Handling:** Write a Java program that performs the following tasks:

- Create a class named `Calculator` with methods for addition, subtraction, multiplication, and division.
- Handle the following exceptions within the `Calculator` class:
 - `ArithmeticException`: Handle division by zero.
 - Custom exceptions: Create at least two custom exceptions, such as `NegativeNumberException` and `OutOfRangeException`, and handle them appropriately.

2. **Main Program:** Create a main program (a separate class) that demonstrates the usage of the `Calculator` class. In the main program, you should:

- Prompt the user to enter two numbers and an operator (+, -, *, /).
- Use the `Calculator` class to perform the specified operation.
- Handle any exceptions that may occur during the calculations and display appropriate error messages.

3. **Custom Exceptions:** Define at least two custom exception classes (e.g., `NegativeNumberException` and `OutOfRangeException`). These should be used within your `Calculator` class to handle specific cases.

4. **Grading Rubric (100 points):**

- **Functionality (40 points):**
 - The program correctly performs addition, subtraction, multiplication, and division.
 - It handles division by zero and custom exceptions appropriately.

- Custom exceptions are defined and used as required.
- **User Interaction (15 points):**
 - The program provides clear and user-friendly prompts and error messages.
 - Input validation is implemented to ensure valid numeric input.
- **Custom Exception Handling (20 points):**
 - At least two custom exceptions are defined.
 - Custom exceptions are correctly thrown and caught.
- **Code Structure and Organization (10 points):**
 - The code is well-organized, readable, and follows Java naming conventions.
 - Proper use of classes, methods, and error handling.
- **Documentation (10 points):**
 - Adequate comments and documentation are provided to explain the code's functionality.
- **Bonus (5 points):**
 - Implement additional functionality (e.g., handling more custom exceptions or extending the calculator) for extra points.

Submission: Submit your Java program files as a compressed archive (ZIP or RAR) along with a brief documentation explaining your custom exceptions, how they are used, and how to run the program.

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 7	Everyone	-	Oct 22 at 11:59pm